

02

/ 13 LECTURES

Network Packets and Layered Communication

Protocols, the OSI & TCP/IP models, encapsulation, frames vs packets vs segments, the Ethernet frame, MTU, fragmentation and MAC addresses.

Protocols

OSI 7 Layers

TCP/IP Model

Encapsulation

Decapsulation

Headers & Payload

Frame · Packet · Segment

Ethernet Frame

MTU

Fragmentation

MAC Addresses

WHAT THIS LECTURE GIVES YOU

How the Internet organises itself into layers — why every message gets wrapped in layer after layer of headers, what each OSI layer is responsible for, exactly how an Ethernet frame is laid out byte-by-byte, what happens when a packet is too big for a link (MTU & fragmentation), and the crucial rule of MAC vs IP scope — with a full worked journey of a web request.

LEARNING OBJECTIVES — AFTER THIS LECTURE YOU CAN

- Define **protocol** and explain **why layered architecture** exists
- Name all **7 OSI layers** and describe each one's responsibility
- Map OSI layers onto the **TCP/IP model** used in practice
- Trace **encapsulation** (down the stack) and **decapsulation** (up the stack)
- Distinguish **frames, packets and segments** — and which addresses each carries
- Read an **Ethernet frame** field-by-field, and explain **MTU** and **fragmentation**
- Describe **MAC address** structure, scope, and how **ARP** connects IP↔MAC

Packets & Layered Communication

Protocol	Layering	OSI — 7 layers	TCP/IP — 4 layers
format · sequence · actions	modularity · reuse · isolation	PDNTSPA mnemonic	what the Internet runs
both sides must agree	each layer = one job	vocabulary for problems	App · Transport · Internet · Net-Access
Encapsulation	Data Units	Ethernet & MTU	MAC & ARP
headers added going down	Segment (L4 · ports)	frame fields byte-by-byte	48-bit · OUI + device
stripped going up	Packet (L3 · IPs)	MTU 1500 B	local scope, changes each hop
	Frame (L2 · MACs)	fragment + reassemble	ARP: IP → MAC

1 What is a Protocol? Why Layers?

A **protocol** is a set of agreed-upon rules governing communication. Both sender and receiver must follow the same protocol or communication fails. In networking a protocol defines three things:

Format

Which fields appear, in what order, what each byte means (the message "grammar").

Sequence

Who initiates, who responds, and in what order (the "conversation flow").

Actions

What to do on events: message lost, corrupted, arrives out of order.

Why layered protocols?

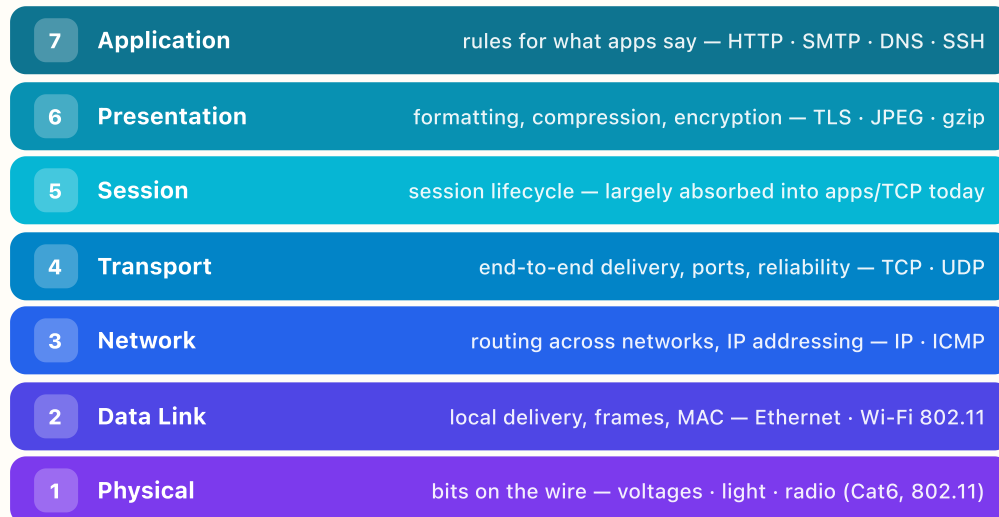
Rather than one massive protocol doing everything, networking uses a **layered architecture**: each layer has **one job**, relies on the layer below, and provides a service to the layer above.

- **Modularity** — change one layer without touching others: Apple can update Safari's HTTP engine without modifying TCP.
- **Reusability** — TCP works identically over Ethernet, Wi-Fi, or 4G.
- **Isolation** — easier diagnosis: if TCP connects but the page is broken, the problem is in the application layer, not the network.

ANALOGY Sending a gift: you (application) write the card; the packing service (transport) boxes it; the courier company (network) routes the box; the local delivery van (data link) carries it on the final street; the road (physical) carries the van. **Each level only cares about its own wrapping.**

2 The OSI Model — 7 Layers

In the 1980s vendors (IBM, DEC...) had incompatible systems. The **ISO** created the **OSI (Open Systems Interconnection) model, 1984**, as a universal reference. Every network function belongs to one of 7 layers:



Layer numbers are used from TOP (7) to BOTTOM (1) — data flows 7→1 on send, 1→7 on receive

Figure 1. The OSI reference model — a common vocabulary, not a running implementation.

Mnemonic (bottom → top)

"Please Do Not Throw Sausage Pizza Away" → Physical, Data Link, Network, Transport, Session, Presentation, Application.

Layer	One-line job	If it breaks... (symptom)
1 — Physical	How bits travel: voltage on copper, light in fiber, radio for Wi-Fi	Cable unplugged — nothing works at all
2 — Data Link	Organises bits into frames ; delivery on ONE network via MAC addresses	Connected but can't talk to devices on the <i>same subnet</i>
3 — Network	Routing across networks via IP addresses	Can reach your router but not the Internet
4 — Transport	Port numbers (right application) + reliability (TCP / UDP)	Connects but data corrupted, lost, or out of order
5 — Session	Open / maintain / close sessions (handled by apps or TCP today)	— (mostly historical)
6 — Presentation	Format translation, compression, encryption — TLS lives here	Connects but content garbled / decryption fails
7 — Application	Protocols apps speak: HTTP, SMTP, DNS	Network fine but server returns errors

3 The TCP/IP Model — What We Actually Run

OSI is a reference; the Internet actually implements the **TCP/IP model** (DARPA, 1970s) with **4 layers**:

OSI		TCP/IP
7 Application	→	Application
6 Presentation		HTTP, DNS, SMTP, SSH...
5 Session		
4 Transport	→	Transport
		TCP, UDP
3 Network	→	Internet
		IP, ICMP
2 Data Link	→	Network Access
1 Physical		Ethernet, Wi-Fi, LTE...

How we use both

We **implement** TCP/IP protocols, but **talk about** them with OSI layer numbers — "that's a Layer 3 problem", "an L2 switch". This course does the same: study TCP/IP protocols, use OSI numbers as vocabulary.

NOTE

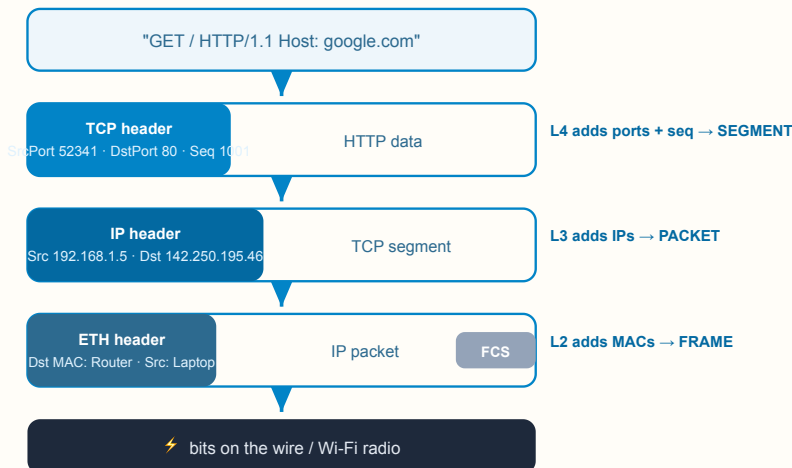
Some textbooks show TCP/IP as 5 layers (Application, Transport, Network, Data Link, Physical) — identical ideas, just with the Network Access layer split apart. Know both counts: **4 or 5**.

4 Encapsulation & Decapsulation

Encapsulation (sender): going **DOWN** the stack, each layer adds its own header around the data from above.

Decapsulation (receiver): going **UP**, each layer strips its header and passes the payload up.

SENDER — encapsulate ↓



RECEIVER — decapsulate ↑

Frame arrives → L2 strips ETH header →
→ L3 strips IP header →
→ L4 strips TCP header →
→ L7 reads the HTTP request ✓

KEY INSIGHT

Each layer treats everything above as opaque payload — just bytes to carry. That's why HTTP can change without touching TCP.

Figure 2. Encapsulation wraps data in layer-headers going down; decapsulation unwraps it going up.

WHY?

Why is the Ethernet destination the router, not Google? Ethernet only works on the *local* network. Your laptop doesn't know (and can't use) Google's MAC address — so L2 delivers to the router, which forwards the IP packet to the next hop and builds a **brand-new Ethernet frame** for that leg.

5 Headers, Payload & the Nesting Rule

HEADER control information

PAYLOAD the actual data being sent

Typical header contents: source & destination addresses (at that layer's scope), payload length, payload type/protocol, sequence numbers, error-check values, control flags (SYN, ACK...).

The nesting rule

Ethernet payload = IP packet → IP payload = TCP segment → TCP payload = HTTP message → HTTP payload = the webpage HTML

6 Frames vs Packets vs Segments

Term	OSI Layer	Protocol	Address type	Scope
Segment	L4 — Transport	TCP / UDP	Port numbers	End-to-end (process → process)
Packet	L3 — Network	IP	IP addresses	Host-to-host (across all networks)
Frame	L2 — Data Link	Ethernet / Wi-Fi	MAC addresses	Local network (one single hop)

What changes at each router hop?

HOP 1 · Laptop → Home Router

ETH: Laptop MAC → Router LAN MAC

IP: 192.168.1.5 → 142.250.195.46

MACs: laptop ↔ router (local)

HOP 3 · ISP → Google

ETH: ISP MAC → Google MAC ← REPLACED

IP: 192.168.1.5 → 142.250.195.46 — same

HOP 2 · Router → ISP Router

ETH: Router WAN MAC → ISP MAC ← REPLACED

IP: 192.168.1.5 → 142.250.195.46 — same

new frame built for this leg

every hop:

MACs change

IPs stay

(until NAT)

Figure 3. The frame is rebuilt at every hop; the IP packet persists end-to-end.

EXAM

The rule: MAC addresses change at every router hop; IP addresses stay the same source→destination (unless NAT — Lecture 11). Expect this as a true/false or short answer.

7 Ethernet Frame — Byte by Byte

8 B	Preamble+SFD	6 B	Dest MAC	6 B	Src MAC	2 B	EtherType	46–1500 B	Payload (usually an IP packet)	4 B	FCS
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Min frame 64 B · Max frame 1518 B Preamble syncs the receiver's clock

Figure 4. Ethernet II frame layout.

- **Preamble (7 B) + SFD (1 B)** — synchronise the receiver's clock; SFD marks "real data starts after me".

- **Destination MAC (6 B)** — **Unicast** one device · **FF:FF:FF:FF:FF:FF** broadcast to all · multicast for groups.
- **Source MAC (6 B)** — sender's hardware address.
- **EtherType (2 B)** — what's inside the payload, so the receiver knows which upper protocol to hand it to.
- **Payload (46–1500 B)** — smaller data is zero-padded up to 46 B (64-byte minimum enables collision detection on old shared Ethernet).
- **FCS (4 B)** — CRC over the whole frame. Mismatch → frame **silently discarded** (no retransmit — recovery is TCP's job, L4).

EtherType	Payload	EtherType	Payload
0x0800	IPv4	0x86DD	IPv6
0x0806	ARP	0x8100	VLAN-tagged frame

8 MTU & IP Fragmentation

MTU — Maximum Transmission Unit: the largest payload a single frame can carry on a link.

Link type	MTU	Link type	MTU
Ethernet	1500 bytes	Jumbo frames (data centre)	up to 9000 B
Wi-Fi (in practice)	1500 B (Ethernet-compatible)	Loopback interface	65536 B

If an IP packet exceeds a link's MTU, the IP layer **fragments** it. Example — 4500-byte payload over Ethernet (MTU 1500):

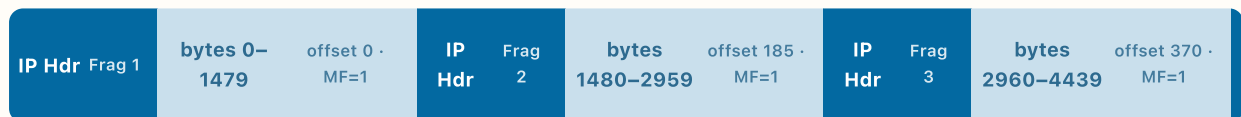


Figure 5. One 4500-byte IP packet → four fragments (1480 bytes of data each + 20-byte header).

IP header field	Purpose in fragmentation
Identification	All fragments of the same original packet share this 16-bit ID
Fragment Offset	Where this fragment starts, in 8-byte units (13-bit field × 8 covers 65,528 B — enough for max IP size 65,535 B)
MF — More Fragments	1 = more pieces coming · 0 = last fragment
DF — Don't Fragment	1 = router must DROP (not fragment) if too large — used by PMTUD

RULES Fragmentation can happen at **any router** whose MTU is smaller — but **reassembly happens ONLY at the final destination**. Lose even one fragment → destination discards all of them → source TCP retransmits the whole thing.

Path MTU Discovery (PMTUD) — how modern systems avoid fragmentation

- 1 **Send with DF = 1**
Packets marked "don't fragment".

2 Small-MTU router drops & reports

It returns ICMP *"Fragmentation Needed, MTU = X"*.

3 Sender adjusts & retries

Repeats until the size fits the whole path — no fragments ever sent.

GOTCHA Firewalls that block **all ICMP** break PMTUD: large transfers (email attachments, image uploads) mysteriously hang while small ones succeed. Classic interview/troubleshooting trap.

9 MAC Addresses & ARP

AA:BB:CC OUI — 3 bytes, IEEE-assigned manufacturer ID

DD:EE:FF Device ID — 3 bytes, unique per NIC

Figure 6. A MAC address = 48 bits, written as six hex pairs: OUI (who made it) + device ID.

FIND YOUR MAC ADDRESS

TERMINAL

```
# Linux / macOS
ip link show      # look for "link/ether"
# Windows
ipconfig /all     # look for "Physical Address"
```

Property	Detail
Scope	Local only — meaningful within one network segment (one hop)
Uniqueness	Globally unique in theory (manufacturer-assigned)
Spoofing	Changeable in software — modern phones randomise MAC per network for privacy
Purpose	Layer-2 delivery identification on the local network

ARP — bridging IP to MAC (preview of Lecture 11)

1 Broadcast question

PC (192.168.1.5) shouts on the LAN (Dst MAC FF:FF:FF:FF:FF:FF): *"WHO HAS 192.168.1.1? TELL 192.168.1.5"*

2 Unicast answer

Router: *"192.168.1.1 is at 11:22:33:44:55:66"*

3 Cache it

PC stores the mapping in its ARP cache and uses that MAC in future frames.

10 Complete Example — One Web Request, Layer by Layer

Scenario: laptop 192.168.1.5 sends GET / to google.com (142.250.195.46).

Step	Layer	What is added / done
1	L7 Application	Browser creates GET / HTTP/1.1 Host: google.com
2	L4 Transport	TCP adds SrcPort 52341 · DstPort 80 · Seq 1001 · ACK
3	L3 Network	IP adds Src 192.168.1.5 · Dst 142.250.195.46 · TTL 64
4	L2 Data Link	Ethernet DstMAC Router · SrcMAC Laptop · EtherType 0x0800 · FCS
5	L1 Physical	NIC transmits as electrical / radio signals
6	At each router	Strip ETH → read Dst IP → routing-table lookup → build new frame for next hop (TTL -1)
7	At Google	Full decapsulation → HTTP request read → response travels back in reverse

LECTURE 2 — RAPID REVISION CARD

Protocol = agreed format + sequence + actions. Layers give modularity, reuse, isolation.

OSI 7: PDNTSPA — Physical, Data Link, Network, Transport, Session, Presentation, Application.

TCP/IP 4: App (5+6+7) · Transport (4) · Internet (3) · Network Access (1+2). Implement TCP/IP, speak OSI.

Encapsulation ↓: headers added each layer going down; **decapsulation ↑:** stripped going up.

Segment L4 ports · **Packet** L3 IPs · **Frame** L2 MACs.

Ethernet frame: Preamble 8 + DstMAC 6 + SrcMAC 6 + EtherType 2 + Payload 46–1500 + FCS 4 = 64–1518 B.

MTU 1500 B; fragmentation at any router; **reassemble only at destination;** one lost fragment → all retransmitted.

MAC 48-bit (OUI+device), local scope, changes every hop; IPs stay end-to-end (until NAT).

ARP: broadcast "who has IP X?" → unicast reply → cache.

EtherType **0x0800** IPv4 · **0x0806** ARP · **0x86DD** IPv6.

Important Questions — Exam Practice

A. Multiple Choice

Q1. Which OSI layer converts packets into frames for local delivery?

MCQ

- a) Network
- b) Data Link
- c) Transport
- d) Physical

Q2. The EtherType value 0x0806 indicates the payload is:

MCQ

- a) IPv4
- b) IPv6
- c) ARP
- d) VLAN tag

Q3. A 4500-byte IP payload over Ethernet (MTU 1500) is split into:

MCQ

- a) 2 fragments
- b) 3 fragments
- c) 4 fragments
- d) 5 fragments

Q4. Where are fragmented IP packets reassembled?

MCQ

- a) At the next router
- b) At every router
- c) Only at the final destination
- d) At the nearest DNS server

Q5. Which fields change at every router hop (no NAT)?

MCQ

- a) Src & Dst IP
- b) Src & Dst MAC and TTL
- c) Ports
- d) EtherType only

B. Short Answer (2–4 marks)

Q6. Define protocol. Why does networking use a layered architecture? (3 benefits)

SHORT

Q7. List the 7 OSI layers in order (both directions) with the mnemonic, and map them to the 4 TCP/IP layers.

SHORT

Q8. Draw and label the Ethernet frame with byte sizes. What is the FCS used for, and what happens on mismatch?

SHORT

Q9. Differentiate frame, packet and segment (layer, address type, scope).

SHORT

Q10. What is MTU? Why does Ethernet enforce a 46-byte minimum payload?

SHORT

Q11. Explain the structure of a MAC address (OUI vs device ID) and its scope. Why do MACs change at each hop while IPs don't?

SHORT

C. Long Answer (5–8 marks)

Q12. Trace encapsulation AND decapsulation for an HTTP GET from laptop (192.168.1.5) to google.com (142.250.195.46), naming every header added and every data-unit name at each layer. Explain why the Ethernet destination is the router's MAC.

LONG

Q13. An IP packet with 4500 bytes of payload crosses an MTU-1500 link. Compute the number of fragments, each fragment's data range, fragment offset (in 8-byte units), and MF bit values. Which IP header fields make reassembly possible?

NUMERICAL

Q14. Explain Path MTU Discovery. Why does a firewall that blocks all ICMP cause "large uploads hang, small ones succeed"?

LONG

Q15. "Each layer treats the layer above as opaque payload." Explain this design principle and how it enables TCP to run unchanged over Wi-Fi, Ethernet or 4G.

LONG

ANSWER KEY

A1. b — Data Link (L2) frames carry packets locally.

A2. c — $0x0806 = \text{ARP}$ ($0x0800 = \text{IPv4}$, $0x86DD = \text{IPv6}$).

A3. c — $\text{ceil}(4500/1480) = 4$ fragments of ≤ 1480 data bytes.

A4. c — reassembly only at the final destination.

A5. b — MACs rewritten each hop; TTL decremented; IPs stay.

A6. Agreed rules: format, sequence, actions. Benefits: modularity, reusability, isolation (any three with one-line explanations).

A7. PDNTSPA; 5+6+7→Application, 4→Transport, 3→Internet, 1+2→Network Access.

A8. Figure 4 layout; FCS = CRC error check; mismatch → silent discard (TCP recovers later).

A9. Segment/TCP/ports/end-to-end · Packet/IP/IPs/host-to-host · Frame/Ethernet/MACs/one hop.

A10. Largest frame payload per link (1500 B Ethernet). 46-byte min → 64-byte min frame for collision detection on legacy shared Ethernet; smaller data is padded.

A11. 48-bit: 3-byte OUI (IEEE, manufacturer) + 3-byte device ID. Local scope only — each hop's frame is freshly built for that local leg; IPs provide the global end-to-end identity.

A13. 4 fragments: 0–1479 (off 0, MF1), 1480–2959 (off 185, MF1), 2960–4439 (off 370, MF1), 4440–4499 (off 555, MF0). Identification + offset + MF enable reassembly.

A12/A14/A15. Use Figures 2 & 5 and the PMTUD steps — graders award labelled diagrams and the "opaque payload" principle stated explicitly.

Wrap-Up — carry these into Lecture 3

ENCAPSULATION

Every layer wraps the layer above: **Data → Segment → Packet → Frame**. Decapsulation reverses it at the receiver.

TWO ADDRESSES

IPs identify hosts **end-to-end** and never change; MACs deliver on the **local link** and are rewritten at every hop.

MTU

1500 B Ethernet limit → packets larger than MTU–20 are **fragmented**; the receiver reassembles using ID + offset + MF.

WHY LAYERS WIN

TCP runs unchanged over Wi-Fi, Ethernet or 4G because each layer treats the one above as **opaque payload**.

NEXT

Lecture 3 — IP Addresses & Subnetting I: we open the IP header's addressing half — octets, network vs host bits, and your first subnet masks.